

Roberto Russo

Principal Software Architect – Energy & Utilities IoT / data platforms

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PROFILE

Principal software architect with 25+ years of engineering experience and, over the last five years, a focus on large-scale real-time IoT and data platforms in the Energy & Utilities domain. Focused on foundational engineering: system architecture, data modeling, frameworks and developer tooling built to a high quality bar, so the rest of the team moves faster, with less friction and higher overall quality. Technical authority for design and for the documentation that anchors it: architecture documents, position papers, design specifications. Domain breadth from renewable generation (wind, photovoltaic, BESS) and smart-city interoperability (PELL) through e-mobility to online gaming. Strong on object-oriented and hexagonal design, predicate and rule-based systems, schema design, and observability architecture; broad polyglot range from C and assembly through game engines to cloud-native. An individual-contributor architect who creates the most value building deep in the engineering, not at the client interface. AI-accelerated delivery; fully remote since 2020.

CORE SKILLS

- **Architecture & design:** conceptual & data modeling, OO design, frameworks & toolkits, reusability, hexagonal / ports-and-adapters, GoF & EIP, technical-debt reduction, zero-warning discipline, systems-level view.
- **Cloud & infra:** AWS (EKS, Kinesis, DynamoDB, Aurora PostgreSQL, cross-account IAM, Pod Identity), Kubernetes, Terraform, CDK / CDKTF, Helm, GitHub Actions, AWS CodeBuild/CodePipeline/CodeDeploy, Jenkins, Git, BitBucket.
- **Data & integration:** MongoDB Atlas, Aurora, DynamoDB, Oracle, SQLite; time-series & streaming pipelines; Apache Camel, Apache Synapse, WSO2 EI / API Manager; Google Protocol Buffers.
- **Languages:** Python; C++11 (Qt, QML, Boost, OpenGL); C# / .NET; Java / JEE; broad eclectic range across paradigms.
- **AI-accelerated delivery:** drove org-wide adoption of AI-assisted development (Claude Code).
- **Documentation & method:** Technical Architecture Documents, position papers, design specifications; Jira, XRay, Confluence, Redmine; Agile.
- **Domains:** Energy & Utilities, Oil & Gas, critical infrastructure, e-mobility; (earlier) online gaming, banking.

EXPERIENCE

Principal Consultant – **Softwave Soluzioni e Tecnologie S.r.l.**

Dec 2015 – Present · Remote

Independent senior consultant delivering solution architecture and senior development to enterprise clients across renewable energy, smart-city, e-mobility and online gaming, on long-term engagements.

Tech Lead / Architect – Plenitude REN, real-time IoT telemetry platform (via Reply · 2021 – Present)

Real-time IoT telemetry platform for Eni Plenitude renewable-energy assets (wind, PV, storage), at the Reply group company whose work earned Reply the AWS Energy & Utilities Partner of the Year recognition.

- designed the end-to-end architecture: edge drivers on EKS, Kinesis streaming ingestion, multi-store persistence (DynamoDB, MongoDB Atlas, Aurora PostgreSQL), and the rule/event engine as the core platform engine;
- built a white-label simulator for reproducible telemetry and test generation bound to real reference data, a reusable cross-project accelerator;
- built a white-label observability (sim+obs) tooling stack monitoring the cloud resources running the solution, a reusable platform asset;
- designed a hexagonal (ports-and-adapters) driver framework: canonical / IoT-gateway / mock adapters; SFTP, SQL and REST drivers; clean extension points for new integrations;
- built a white-label plant-onboarding toolkit (transactional idempotent loader, gated fail-closed rule-seeder, local overlay UI, hexagonal vendor adapter): reusable across plants and clients;
- built workstation-to-cloud developer tooling, reducing build and release friction for the team;
- ran technical design reviews and written analyses that raised the overall quality bar;
- drove org-wide adoption of AI-assisted development, turning individual practice into a unit-wide productivity gain.

Stack: Python, AWS (EKS, Kinesis, DynamoDB, Aurora, IAM), Kubernetes, MongoDB Atlas, Terraform, Helm.

Enel Green Power – ROMAP migration to the new Enel Platform (via Reply · 2021 – 2023)

Migration of the ROMAP application onto the new microservices-based Enel Platform. ROMAP supports drone inspection campaigns on energy assets (wind turbines, PV strings): acquisition and classification of photographic and thermographic imagery, and maintenance-backlog scheduling. Re-architecture of monolithic features into cloud-native microservices.

Other enterprise engagements (via Reply • 2021+)

Enel X, Q8 Italia, Aeroporti di Roma: solution architecture and senior development across e-mobility, oil & gas and infrastructure.

Senior Software Developer – Sisal (Softwave engagement • 2019 – 2021)

B2B and B2C core betting platform development.

Stack: Java EE, Oracle Coherence, Apache Kafka, Spring Boot, REST, Docker, Ansible, Git.

Software Developer / DevOps – Eurobet Italia (Softwave engagement • 2018 – 2019)

Microservice-oriented core platform design and development; development life-cycle design: CI/CD on Jenkins, functional test automation (Behave, Selenium), containerization.

Stack: UML, C++17, Redis, gRPC, RestSDK, POSIX IPC, Linux, Oracle, Git, Docker.

Software Developer – IGT

Feb 2012 – Apr 2018

Client-side and server-side design and development for online multiplayer gaming.

- re-engineering of the online-poker real-time fraud-detection system based on player-behaviour analysis;
- stack: C++11, Qt, QML, Boost, OpenGL, JEE, Python, C#, .NET, UML, Google Protocol Buffers, Oracle, SQLite, Jenkins, Subversion.

EARLIER EXPERIENCE

Software Developer – Banca Popolare di Sondrio (via system integrator) • Jul 2009 – Dec 2011

Middleware design and B2B communication integration between a travel-services agency and Banca Popolare di Sondrio for payment automation over SOA infrastructure, compliant with bank-to-bank security standards. Real-time bank reporting and accounting-system re-engineering.

Stack: relational modeling, Java, Apache Axis, TLS/SSL, IKVM, .NET, WCF, C#, LINQ, ASP.NET, NLog, Microsoft SQL Server.

Instructor – Italian Air Force (Aeronautica Militare) • Feb 2009 – Jun 2009

Five-month full-time "Object-Oriented Design" course at the Italian Air Force Headquarters, DIPMA, Rome. Lectures plus on-the-job training.

Consultant – Regione Lazio (regional vocational-counseling agency) • 2002 – 2009

Design and development of a career-exploration and vocational-counseling toolkit: computer-aided administration and analysis of consolidated psychometric tests (Holland 85, MBTI, RIASEC), interactive 3D visualization via statistical MDS (multidimensional scaling), scenario navigation.

Stack: relational modeling, data mining, machine learning, Orange, WEKA, SAP PowerDesigner, SQL, C#, ASP.NET, haXe, Flash.

Software Developer – Sapienza Università di Roma • 2007 – 2008

Research-related software development.

Self-employed – Consultant, developer, instructor • 1996 – 2002

Independent consulting, software development, and teaching engagements.

EDUCATION

- Sapienza Università di Roma – Computer Science, Faculty of Mathematical and Physical Sciences • 1996 – 2000;
- Convitto Nazionale Vittorio Emanuele II – Scientific High School (Liceo Scientifico) • 1987 – 1992.

LANGUAGES

- Italian – native;
- English – strong written and reading proficiency (technical documentation, design, async collaboration); conversational spoken;
- Spanish – conversational fluency; good listening comprehension.

INTERESTS

Software craftsmanship and engineering culture; human factors of software engineering; information security (pentesting, computer forensics, cryptography); distributed large-scale computing.